Network Design Report

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Network Topology & Design

The network consists of two autonomous systems separated by a Service Provider (SP) cloud link. Cairo forms AS 100 and Alexandria forms AS 200. Within Cairo, two buildings each have a dedicated access router (R1 for Building 1, R2 for Building 2) that connect upward to a central border gateway router (R3). R3 handles both internal EIGRP routing and external BGP peering with Alexandria's router (R4).

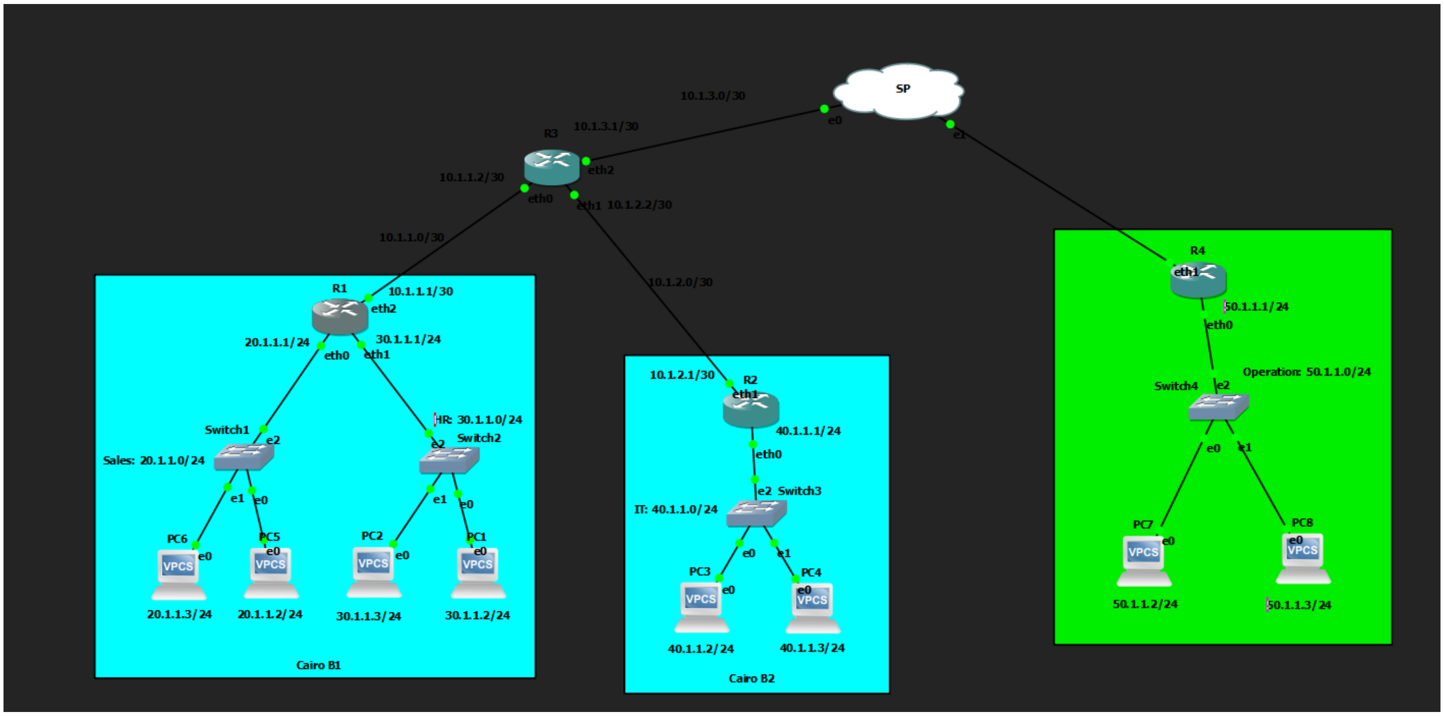
2.1 Topology Overview

Device	Role	Location	Connected To
R1	Access Router	Cairo B1	SW1 (Sales), SW2 (HR), R3
R2	Access Router	Cairo B2	SW3 (IT), R3
R3	Border Gateway	Cairo Hub	R1, R2, SP Cloud (BGP)
R4	Border Gateway	Alexandria B1	SW4 (Operation), SP Cloud (BGP)
SW1–SW4	Ethernet Switch	Per building	Department PCs + access router

Routing Architecture

The routing design uses a two-layer approach:

- Internal layer: EIGRP AS 100 runs between R1, R2, and R3 within Cairo. EIGRP AS 200 runs between R4 and its internal networks in Alexandria.
- External layer: eBGP runs between R3 (AS 100) and R4 (AS 200) over the SP cloud link using the 10.1.3.0/30 point-to-point subnet.
- Redistribution: R3 redistributes EIGRP routes into BGP (so Cairo networks are advertised externally) and redistributes BGP routes back into EIGRP (so internal Cairo routers learn the Alexandria route). R4 performs the same mutual redistribution on the Alexandria side.


SP
10.1.3.0/30
10.1.3.1/30
R3
eth2
e0
e1
10.1.1.2/30
eth0
eth1
10.1.2.2/30
10.1.1.0/30
10.1.2.0/30
R4
eth1
50.1.1.1/24
eth0
Operation: 50.1.1.0/24
Switch4
e2
e0
e1
PC7
PC8
50.1.1.2/24
50.1.1.3/24
R1
10.1.1.1/30
eth2
20.1.1.1/24
30.1.1.1/24
eth0
eth1
Switch1
Sales: 20.1.1.0/24
HR: 30.1.1.0/24
Switch2
PC6
PC5
PC2
PC1
VPCS
20.1.1.3/24
20.1.1.2/24
30.1.1.3/24
30.1.1.2/24
Cairo B1
10.1.2.1/30
R2
40.1.1.1/24
Switch3
IT: 40.1.1.0/24
PC3
PC4
40.1.1.2/24
40.1.1.3/24
Cairo B2

3. IP Addressing Scheme

All department networks use Fixed Length Subnet Masking (FLSM) with /24 prefix length, providing 254 usable host addresses per subnet — sufficient for the 2-PC departments specified in the project brief. Router interconnect links use /30 subnets (2 usable hosts) to conserve address space on point-to-point links.

3.1 Department Networks (LAN Subnets)

Branch	Building	Department	Network	Gateway	PCs
Cairo	B1	Sales (SW1)	20.1.1.0/24	20.1.1.1 (R1 eth0)	PC5, PC6
Cairo	B1	HR (SW2)	30.1.1.0/24	30.1.1.1 (R1 eth1)	PC1, PC2
Cairo	B2	IT (SW3)	40.1.1.0/24	40.1.1.1 (R2 eth0)	PC3, PC4
Alexandria	B1	Operation (SW4)	50.1.1.0/24	50.1.1.1 (R4 eth0)	PC7, PC8

3.2 Router Interconnect Links (/30 Point-to-Point)

Link	Subnet	Router A IP	Router B IP
R1 ↔ R3	10.1.1.0/30	10.1.1.1 (R1 eth2)	10.1.1.2 (R3 eth0)
R2 ↔ R3	10.1.2.0/30	10.1.2.1 (R2 eth1)	10.1.2.2 (R3 eth1)
R3 ↔ R4 (SP)	10.1.3.0/30	10.1.3.1 (R3 eth2)	10.1.3.2 (R4 eth1)

3.3 End Device IP Assignments

PC	Department	IP Address	Default Gateway	Subnet
PC6	Sales	20.1.1.3	20.1.1.1	/24
PC5	Sales	20.1.1.2	20.1.1.1	/24
PC1	HR	30.1.1.2	30.1.1.1	/24
PC2	HR	30.1.1.3	30.1.1.1	/24
PC3	IT	40.1.1.2	40.1.1.1	/24
PC4	IT	40.1.1.3	40.1.1.1	/24
PC7	Operation	50.1.1.2	50.1.1.1	/24
PC8	Operation	50.1.1.3	50.1.1.1	/24

4. Routing Protocol Configuration

4.1 EIGRP — Internal Cairo Routing (AS 100)

EIGRP was chosen as the internal routing protocol for the Cairo branch. It is a Cisco-enhanced distance-vector protocol that supports fast convergence and classless routing. All three Cairo routers participate in EIGRP AS 100.

R1 Configuration

```
interface eth0
  ip address 20.1.1.1/24
  no shutdown
interface eth1
  ip address 30.1.1.1/24
  no shutdown
interface eth2
  ip address 10.1.1.1/30
  no shutdown
router eigrp 100
  network 20.1.1.0/24
  network 30.1.1.0/24
  network 10.1.1.0/30
```

R2 Configuration

```
interface eth0
  ip address 40.1.1.1/24
  no shutdown
interface eth1
  ip address 10.1.2.1/30
  no shutdown
router eigrp 100
  network 40.1.1.0/24
  network 10.1.2.0/30
```

4.2 BGP — Inter-Branch Routing

BGP (Border Gateway Protocol) was used to connect the two autonomous systems across the SP cloud. R3 (AS 100) peers with R4 (AS 200) using eBGP over the 10.1.3.0/30 link. Mutual redistribution bridges the EIGRP and BGP domains on both border routers.

R3 Configuration (Cairo Border Gateway)

```
interface eth0  ->  10.1.1.2/30  (to R1)
interface eth1  ->  10.1.2.2/30  (to R2)
interface eth2  ->  10.1.3.1/30  (to SP / R4)
router eigrp 100
  network 10.1.1.0/30
  network 10.1.2.0/30
```

```
  redistribute bgp
router bgp 100
  no bgp ebgp-requires-policy
  neighbor 10.1.3.2 remote-as 200
  address-family ipv4 unicast
    redistribute eigrp
```

R4 Configuration (Alexandria Border Gateway)

```
interface eth0  ->  50.1.1.1/24  (to SW4 / Operation dept)
interface eth1  ->  10.1.3.2/30  (to SP / R3)
router eigrp 200
  network 50.1.1.0/24
  redistribute bgp
router bgp 200
  no bgp ebgp-requires-policy
  neighbor 10.1.3.1 remote-as 100
  address-family ipv4 unicast
    redistribute eigrp
```

4.3 Key Design Decisions

Decision	Rationale
EIGRP for internal routing	Fast convergence, supports classless subnets, easy configuration in FRR
BGP for inter-AS routing	Industry standard for inter-AS routing; mandatory when crossing autonomous system boundaries
/30 on router links	Minimises address waste on point-to-point links (only 2 hosts needed)
/24 on LANs (FLSM)	Provides 254 host addresses per subnet, simple to manage, meets project FLSM requirement
Mutual redistribution	Allows EIGRP routes to be advertised via BGP externally, and BGP-learned routes to be injected into EIGRP internally
Ethernet Switch as SP	GNS3 Cloud node cannot forward between internal ports; Ethernet Switch correctly passes traffic between R3 and R4

5. Verification & Testing

The following verification steps confirm that the network is fully operational end-to-end.

5.1 Verification Commands

Command	Run On	Expected Result
`show ip bgp summary`	R3 and R4	State/PfxRcd shows a number (not Active), confirming BGP session is established
`show ip route`	R1 / R2	50.1.1.0/24 appears as D EX (External EIGRP redistributed from BGP)
`show ip route`	R4	20.1.1.0/24, 30.1.1.0/24, 40.1.1.0/24 appear as D EX routes
`ping 10.1.3.2`	R3	100% success — confirms L3 connectivity across the SP switch
`ping 50.1.1.3`	PC6 (Sales)	Successful ping — full end-to-end path Cairo Sales to Alexandria Operation

5.2 Troubleshooting Encountered

Issue	Root Cause	Resolution
BGP stuck in Active state	GNS3 Cloud node used as SP — cannot forward between internal ports	Replaced Cloud node with Ethernet Switch, which forwards automatically
BGP shows (Policy) — no prefixes exchanged	FRR requires explicit route policy; blocks all prefixes by default on eBGP sessions	Added 'no bgp ebgp-requires-policy' on both R3 and R4
Destination unreachable from PC	50.1.1.0/24 missing from R1 routing table due to BGP not being established	Fixed by resolving BGP issues above; route appeared automatically after redistribution

6. Conclusion

The Company X network was successfully designed and implemented, connecting four departments across two branches in Cairo and Alexandria. The design meets all project requirements:

- FLSM /24 subnets assigned to each department with correct gateway configuration
- EIGRP AS 100 provides fast, reliable internal routing within the Cairo branch
- BGP eBGP peering between AS 100 and AS 200 enables inter-branch communication
- Mutual redistribution on R3 and R4 ensures all routes are propagated correctly

- Full end-to-end reachability verified by pinging across both branches

The key lesson from implementation was that GNS3's Cloud node is designed for bridging to real network interfaces and cannot be used as an SP simulation node. Replacing it with an Ethernet Switch resolved the BGP connectivity issue. Additionally, FRR's default policy-based filtering required the 'no bgp ebgp-requires-policy' command to allow route exchange without explicit route maps.